Free and Open Source Platforms for Robotic Simulations — Table 1

Robotic simulation platform	URL
Webots	cyberbotics.com
Gazebo	gazebosim.org
Robot Operating System	ros.org
Open Robot Control Software	orocos.org
Yet Another Robot Platform	yarp.it/git-master
Mobile Robot Programming Toolkit	mrpt.org
Robotarium	robotarium.gatech.edu
Poppy-Project	poppy-project.org
CoppeliaSim	coppeliarobotics.com

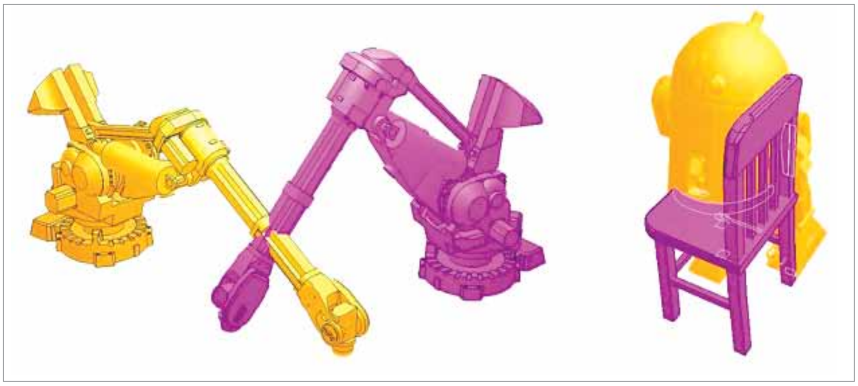

Fig. 3: Collision detection in Coppelia robotic platform

sciences, and many others:

- Remote APIs
- Collision detection
- Minimum distance calculation
- Dynamics/Physics
- Dynamic particles
- Kinematics
- Proximity sensor simulation
- Vision sensor simulation
- Data recording and visualisation
- Building block concept
- Path and motion planning
- Integrated editing modes for customisation
- Custom user interfaces
- Convenient and effective model browser
- High performance data import/export
- RRS interface and motion library
- Full interaction modes in multiple dimensions and coordinates
- Full-featured scene hierarchy

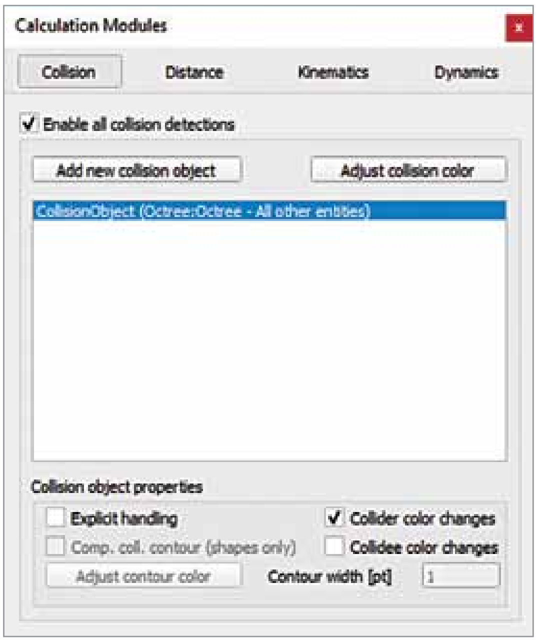

Fig. 4: Setting panel to detect distance for collision avoidance

The implementations, including the convergence of dynamics and physics, can be achieved using CoppeliaSim with Bullet Mechanics, Newton, Differential Equations, and other integrations for easy and effective dynamic calculations to simulate real-life physical and entity encounters. These include collision reaction, grabbing, collision response, grasping, and many others. Multiple recordable data sources (including user-dependent) can view or combine time graphs with xy-graphs or 3D curves that can be achieved effectively.

CoppeliaSim integrates features for real time automations including next to proximity sensors, vision sensors, and viewable objects for dynamic environment with camera objects.

Collision detection and avoidance

Following the brief description of implementation associated with the collision detection and avoidance is one of the key features in robotics. For industrial and corporate applications, if a robot is developed, that robot should be programmed in such a way that it would not collide with other components or machinery in the industrial unit.

The collision detection and avoidance module are easy to program in CoppeliaSim so that the look-and-feel of the robotic application will be there. Through Menu→Tools→Calculation the distance and collision settings can be set up effectively.

The research scholars, scientists, and practitioners have a huge scope to simulate their research tasks using these open source platforms for robotic applications where actual infrastructure and devices are very costly to implement. Rather than using the actual hardware and gadgets, the open source libraries and frameworks provide the ease to researchers to create, simulate, and fetch the research based outcomes from their algorithms and imaginations. EFY

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